

EEE F313: ADVD Lab 4 Report

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1 Introduction

This lab had the goal of creating a 2-input NAND and NOR gate using CMOS technology. We then used the create cellview option to use these modules to build a basic SR latch and test it.

2 2-Input NAND Gate

Using basic CMOS implementation, a 2-input NAND gate was designed. The inputs were designated as A and B , and the output as Y .

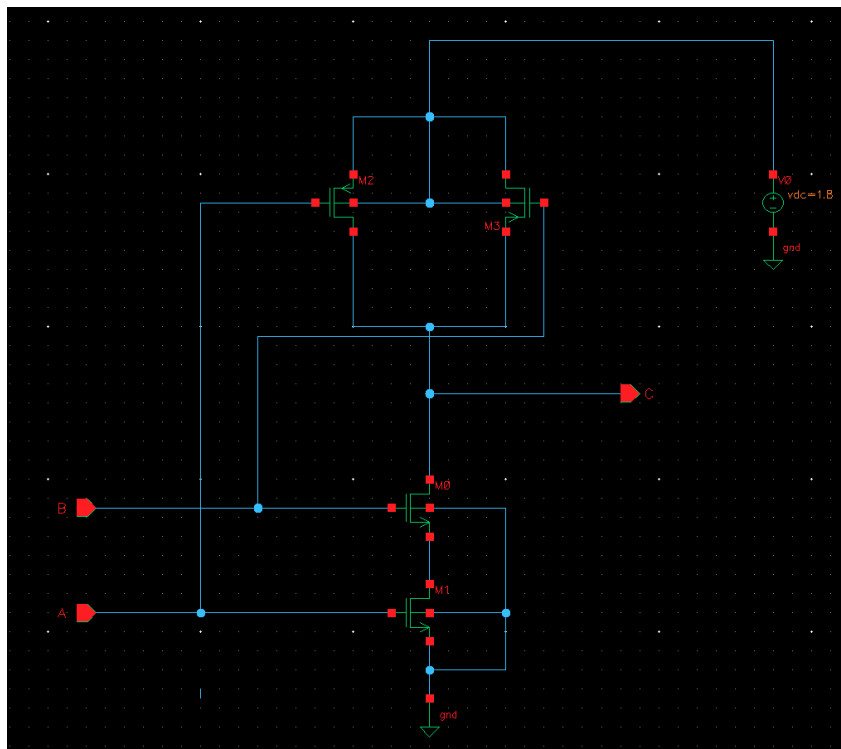


Figure 1: 2 Input NAND Gate using CMOS Technology

This circuit was then exported as a cellview to be used in the SR latch.

3 2-Input NOR Gate

Using basic CMOS implementation, a 2-input NOR gate was designed. The inputs were designated as A and B , and the output as Y .

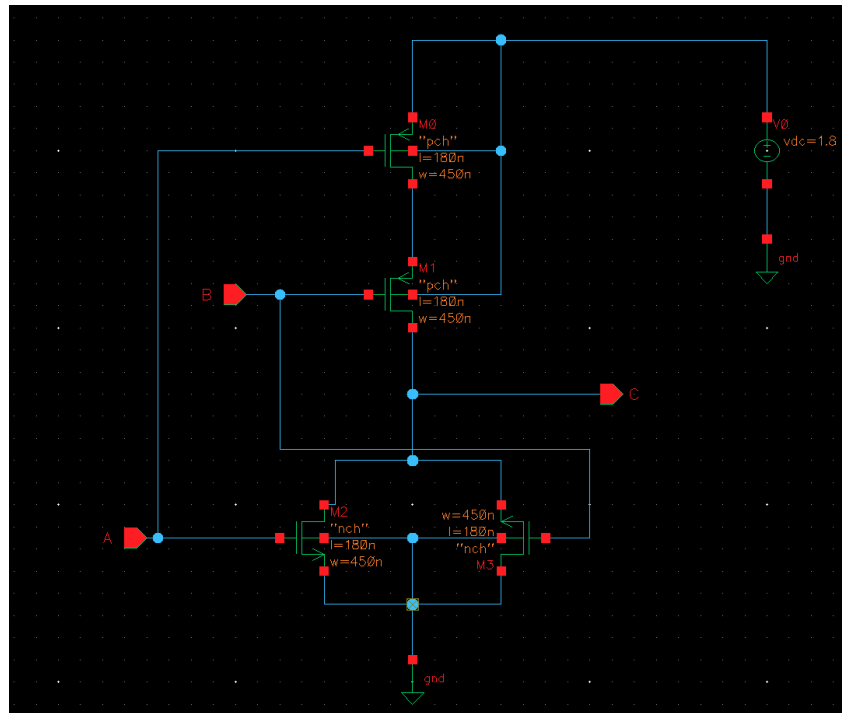


Figure 2: 2 Input NOR Gate using CMOS Technology

This circuit was then exported as a cellview to be used in the SR latch.

4 SR Latch using NAND Gates

An SR latch was designed using the previously created 2-input NAND gate cellview. The inputs were designated as S and R , and the outputs as Q and $\bar{Q}$.

For brevity's sake, only the NAND implementation is shown, but both the NAND and NOR implementations worked.

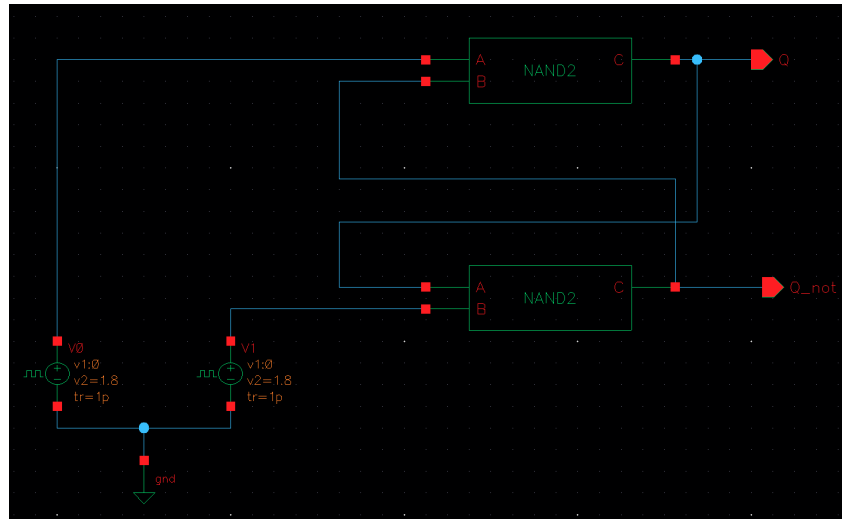


Figure 3: SR Latch Circuit using 2-input NAND Gates

Figure 3 shows the schematic of the SR latch using NAND gates.

A transient analysis was performed using the oscillating voltage sources shown in Figure 3 to test the functionality of the SR latch. The results are shown in Figure 4.

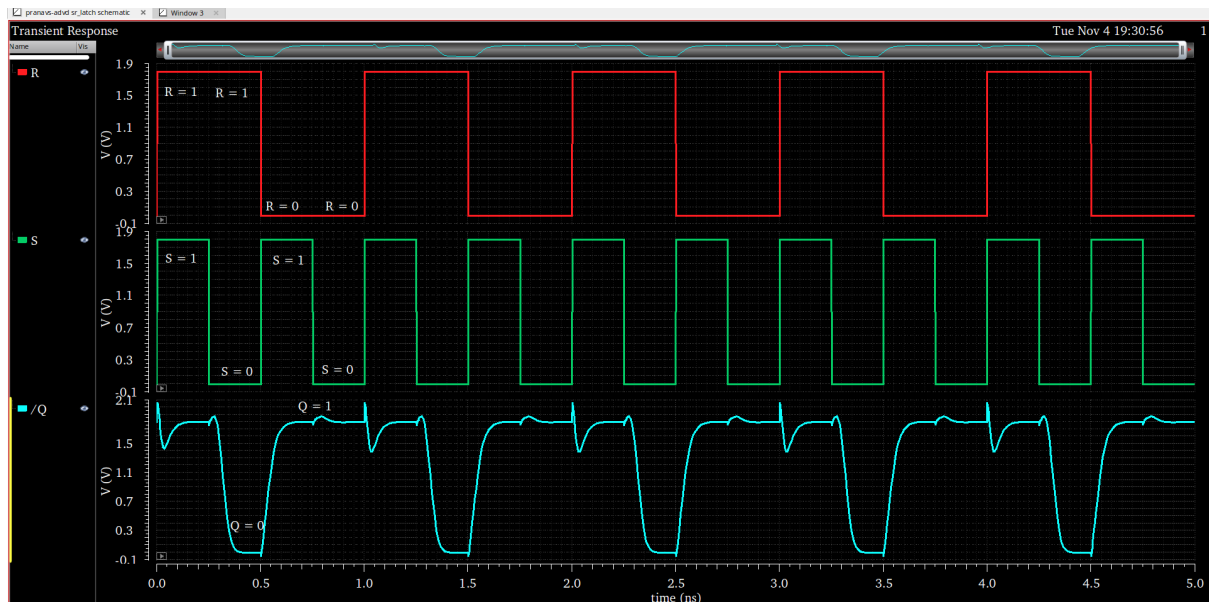


Figure 4: Transient Analysis of SR Latch using NAND Gates